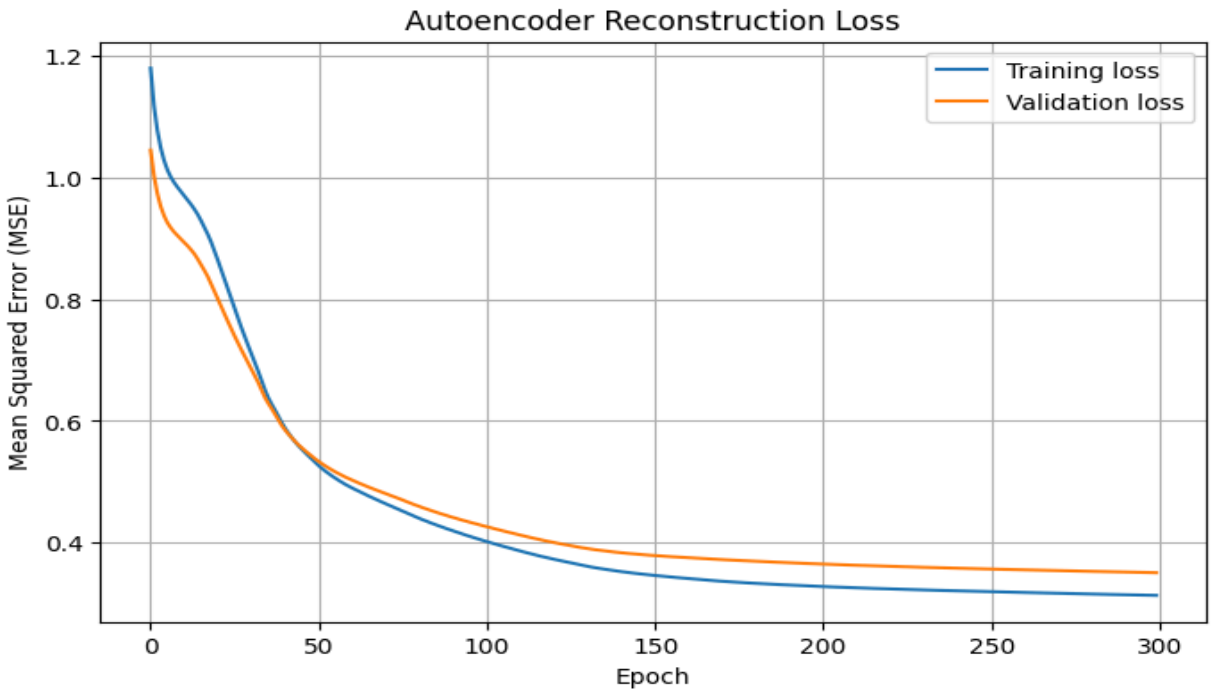
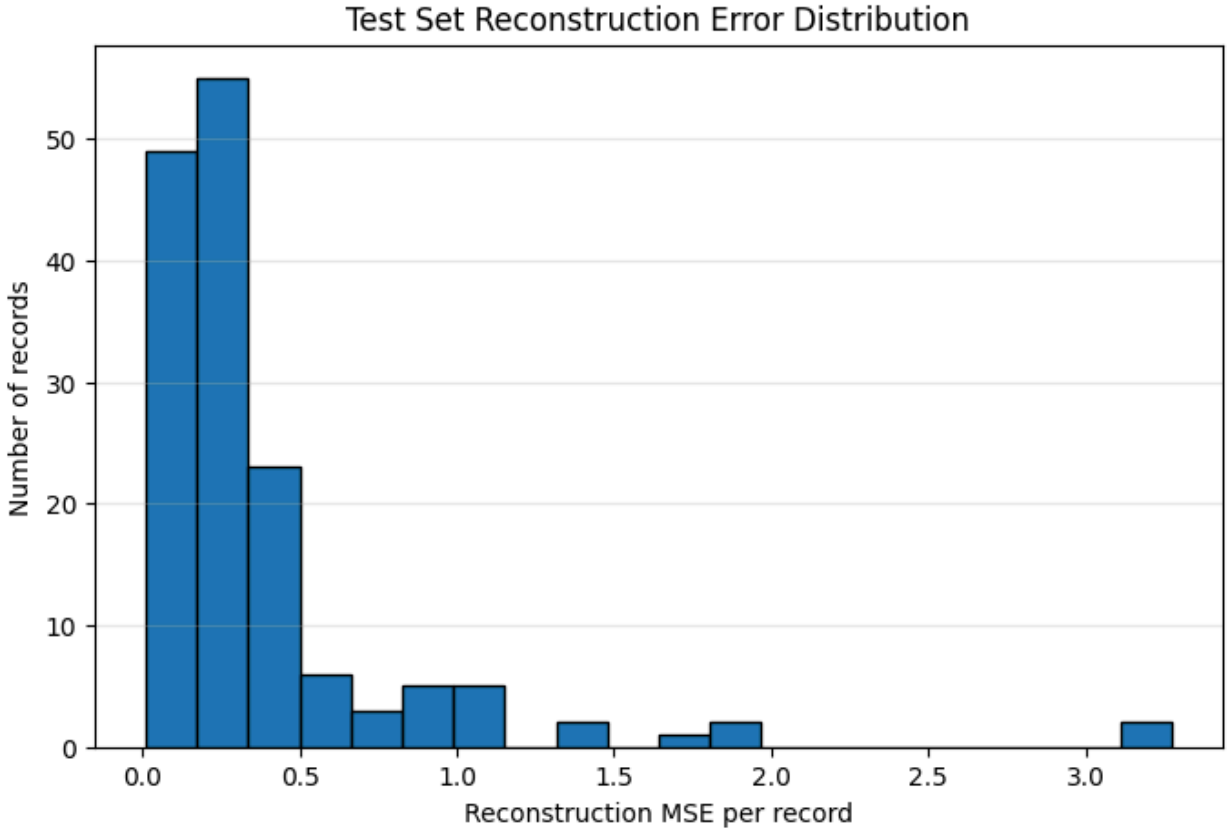
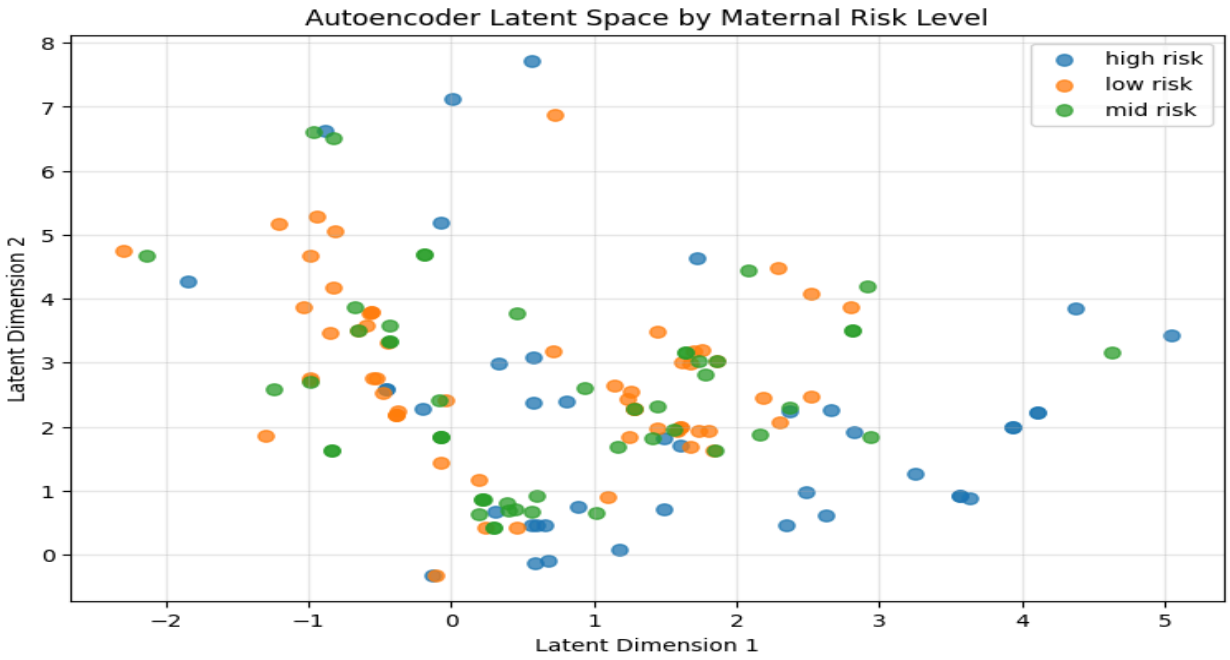
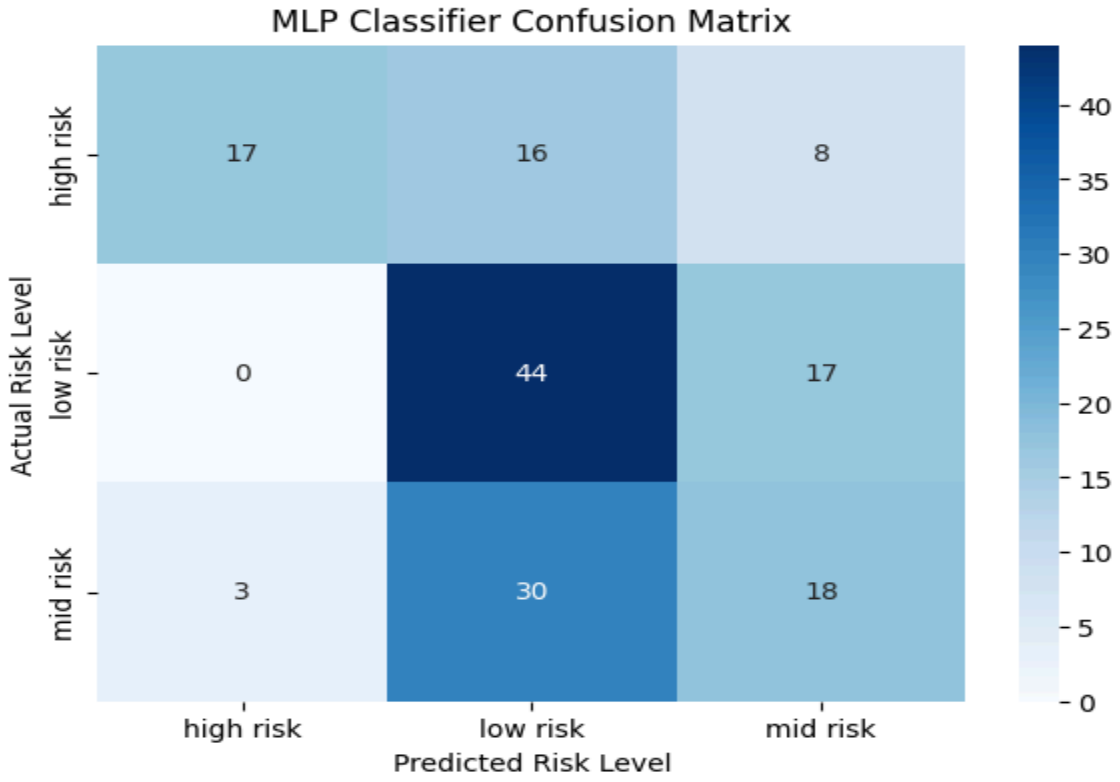

Autoencoder Reconstruction Loss
Training loss
Validation loss
Mean Squared Error (MSE)
1.2
1.0
0.8
0.6
0.4
0
50
100
150
200
250
300
Epoch


Test Set Reconstruction Error Distribution
Number of records
50
40
30
20
10
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Reconstruction MSE per record


Autoencoder Latent Space by Maternal Risk Level
high risk
low risk
mid risk
Latent Dimension 2
Latent Dimension 1
8
7
6
5
4
3
2
1
0
−2
−1
0
1
2
3
4
5


MLP Classifier Confusion Matrix
Actual Risk Level
high risk
low risk
mid risk
17
16
8
0
44
17
3
30
18
high risk
low risk
mid risk
Predicted Risk Level
40
35
30
25
20
15
10
5
0

Majority-class baseline comparison

	Model	Accuracy	Macro F1	Weighted F1
0	Majority-Class Baseline	0.3987	0.1900	0.2273
1	Autoencoder + MLP (SGD)	0.5163	0.5077	0.5094